

The character/root-of-unity ceiling for the parity-twisted descent statistic $s(T)$

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Abstract

This is a negative-results note, with one conjecture. I study the parity-twisted descent statistic $s(T)$ attached to standard Young tableaux, which I have conjectured (“Theorem B”) to record the integer-graded Newton-polygon valuations of a Baxterised monodromy on the irreducible S_n -module V^λ . The question here is sharp and structural: *how much of $s(T)$ can the classical character / root-of-unity machinery see?* Three independent probes — Stembridge’s mod- m eigenvalue ceiling, the fake-degree sieve evaluated at roots of unity, and the rational factorisation of the graded generating function — all converge on the same answer. The machinery sees only cyclotomic-evaluation *shadows* of $s(T)$: its sign $(-1)^{s(T)}$, certain character values, and a single 2-core vanishing order. It cannot see $s(T)$ as an integer grading. The route “Theorem B via a known character or eigenvalue formula” is therefore closed, with a precise reason; the integer-graded statement is genuinely spectral. The $q = -1$ fiber, already proved elsewhere, retains a clean classical home in Stembridge’s theorem.

Setup and definitions

Throughout, $\lambda \vdash n$ is a partition of n , and $\text{SYT}(\lambda)$ denotes the set of standard Young tableaux of shape λ .

Definition 1. For $T \in \text{SYT}(\lambda)$, the *descent set* is

$$\text{Des}(T) = \{i \in \{1, \dots, n-1\} : i+1 \text{ lies strictly below } i \text{ in } T\},$$

and we write

$$\text{maj}(T) = \sum_{i \in \text{Des}(T)} i, \quad \text{comaj}(T) = \sum_{i \in \text{Des}(T)} (n-i).$$

Definition 2 (Parity-twisted descent statistic). For $T \in \text{SYT}(\lambda)$ set

$$s(T) = \sum_{i \in \text{Des}(T)} w_i, \quad w_i = \begin{cases} 2i-1, & n-i \text{ odd,} \\ 0, & n-i \text{ even.} \end{cases}$$

We call a descent *i free* when $w_i = 0$ (i.e. $n - i$ even) and *paid* when $w_i = 2i - 1$ (i.e. $n - i$ odd).

Recall the *fake degree*

$$f^\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{\text{maj}(T)} = (q; q)_n s_\lambda(1, q, q^2, \dots), \quad (q; q)_n = \prod_{r=1}^n (1 - q^r),$$

the second equality being Stanley's principal specialisation of the Schur function. We write

$$G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$$

for the generating function of the parity-twisted statistic.

Context (prior work, not reproved here). In earlier notes I conjectured *Theorem B*: the multiset $\{s(T) : T \in \text{SYT}(\lambda)\}$ coincides with the multiset $\{d_j\}$ of eigenvalue-collision orders — the Newton-polygon valuations as $x \rightarrow q^2$ — of a Baxterised monodromy operator $M(x)$ acting on V^λ . That is the *spectral* side. The $q = -1$ fiber of this picture, the identity

$$\sum_{T \in \text{SYT}(\lambda)} (-1)^{s(T)} = \chi^\lambda(w_0),$$

with w_0 the longest element (reversal), is already proved, via an evacuation / principal-specialisation / Frobenius / fake-degree-sieve argument. The present note does not touch the spectral side. It instead delimits, with rigorous upper bounds where possible and honest computational scope otherwise, exactly what the classical character and root-of-unity machinery can resolve about the *integer-graded* statistic $s(T)$.

1 The Stembridge ceiling

The natural classical home for “a statistic read off a representation” is the eigenvalue distribution of a group element acting on V^λ . Stembridge's theorem makes this precise and, as we shall see, also makes precise its limitation.

Theorem 1 (Stembridge 1989, Thm. 3.3). *Let $w \in S_n$ have cycle type $\mu = (\mu_1, \dots, \mu_\ell)$ and order $m = \text{lcm}(\mu)$. Partition $\{1, \dots, n\}$ into consecutive blocks of sizes $\mu_1, \mu_2, \dots$ in order, and for k in a block of size μ_j , let i be the position of k within its block (so $1 \leq i \leq \mu_j$) and set*

$$b_\mu(k) = i \cdot \frac{m}{\mu_j}.$$

Define the μ -index $\text{ind}_\mu(T) = \sum_{k \in \text{Des}(T)} b_\mu(k) \pmod{m}$. Then, with $\omega = e^{2\pi i/m}$, the multiset of eigenvalues of the representing matrix $\rho_\lambda(w)$ on V^λ is

$$\{\omega^{\text{ind}_\mu(T)} : T \in \text{SYT}(\lambda)\}.$$

Two companion facts frame this. First, Stembridge’s Proposition 1.1: the generating polynomial $P_{\chi,g}(q)$ recording these eigenvalue-exponents is well defined only modulo $1 - q^m$ — the exponents live in $\mathbb{Z}/m\mathbb{Z}$, not in $\mathbb{Z}$. Second, Theorem 3.2: at the regular element of order $m = n$ (the long cycle direction) the indices specialise to the *generalised exponents*, recovering the genuine integer statistic maj. These are the only two regimes the framework offers: integer grading exactly at $m = n$, and otherwise a residue in $\mathbb{Z}/m\mathbb{Z}$.

Now specialise to the involution w_0 , whose cycle type is $\mu = (2^{\lfloor n/2 \rfloor}, 1^{n \bmod 2})$, of order $m = 2$.

Proposition 1 (Stembridge ceiling for $s(T)$). *For w_0 and n even, the Stembridge weights reduce mod 2 to the parity-twist weights term by term:*

$$b_\mu(k) \equiv w_k \pmod{2} \quad (k = 1, \dots, n-1),$$

both equal to the alternating pattern $(1, 0, 1, 0, \dots)$. Consequently

$$\text{ind}_\mu(T) \equiv s(T) \pmod{2} \quad \text{for all } T \in \text{SYT}(\lambda),$$

so the eigenvalue multiset of $\rho_\lambda(w_0)$ is $\{(-1)^{s(T)} : T \in \text{SYT}(\lambda)\}$, whose sum is $\chi^\lambda(w_0)$. Hence the eigenvalue-of-an-element framework recovers exactly the residue $s(T) \bmod 2$ and no more.

Proof. The blocks of w_0 for n even are the $n/2$ pairs $\{1, 2\}, \{3, 4\}, \dots, \{n-1, n\}$, each of size $\mu_j = 2$, so $m/\mu_j = 1$ and $b_\mu(k) = i$ with $i \in \{1, 2\}$ the position of k inside its pair. Thus $b_\mu(k) = 1$ for k odd and $b_\mu(k) = 2 \equiv 0$ for k even, i.e. $b_\mu(k) \equiv 1$ iff k is odd. On the other side, $w_k = 2k - 1 \neq 0$ iff $n - k$ is odd; with n even this is iff k is odd, and then $w_k = 2k - 1$ is odd. So $w_k \equiv 1 \pmod{2}$ iff k is odd as well. The two weight patterns agree mod 2, term by term, giving $\text{ind}_\mu(T) \equiv s(T) \pmod{2}$. The eigenvalues are then $\omega^{\text{ind}_\mu(T)} = (-1)^{s(T)}$, and their sum is $\text{tr } \rho_\lambda(w_0) = \chi^\lambda(w_0)$. $\square$

Remark 1. This is a genuine *ceiling*, not merely a coincidence at w_0 . By Stembridge’s Proposition 1.1, the only element whose eigenvalue exponents are honest integers (rather than residues mod m) is the order- n regular element, and there the statistic produced is maj, which agrees with $s(T)$ only mod 2 — indeed maj and s have entirely different distributions in general. No single group element, and no $\mathbb{Z}$ -linear combination of the functionals “eigenvalue distribution of $\rho_\lambda(w)$,” returns the integer $s(T)$. The character / eigenvalue machinery is intrinsically modular; $s(T)$ as an integer grading is not in its image.

The conclusion of this section is therefore a structural pruning: *Theorem B does not reduce to an eigenvalue or character computation.* The integer grading $G_\lambda(q)$ carries information strictly finer than any $\rho_\lambda(w)$ spectrum, and that information must be sought on the spectral side — the Newton polygon of $M(x)$ — not in the character table.

2 The fake-degree sieve at roots of unity

A second, independent probe is to evaluate the fake degree itself at roots of unity. This is where the $q = -1$ fiber acquires its clean classical home, and it shows exactly how that cleanliness degrades at other roots.

Expanding s_λ in power sums and using $f^\lambda = (q; q)_n s_\lambda$ gives the sieve

$$f^\lambda(q) = \sum_{\mu \vdash n} z_\mu^{-1} \chi^\lambda(\mu) g_\mu(q), \quad z_\mu = \prod_r r^{m_r} m_r!,$$

$$g_\mu(q) = (q; q)_n \prod_{r \geq 1} (1 - q^r)^{-m_r(\mu)},$$

where $m_r(\mu)$ is the number of parts of μ equal to r .

Vanishing/pole bookkeeping at ζ_d . Let ζ_d be a primitive d -th root of unity, $2 \leq d \leq n$. The factor $(q; q)_n = \prod_{r=1}^n (1 - q^r)$ vanishes to order $\#\{r \leq n : d \mid r\} = \lfloor n/d \rfloor$ at $q = \zeta_d$. The product $\prod_r (1 - q^r)^{-m_r(\mu)}$ has a pole there of order equal to the number of parts of μ divisible by d . Since each such part is at least d , the number of parts divisible by d is at most $\lfloor n/d \rfloor$, with equality exactly when μ is *richest* in parts divisible by d (greedily packing multiples of d). Thus $g_\mu(\zeta_d) = \lim_{q \rightarrow \zeta_d} g_\mu(q)$ is finite and nonzero only for the richest classes; all others contribute 0 in the limit.

Proposition 2 (Clean evaluation at a unique richest class). *Suppose the richest-in-parts- $\div d$ class is unique, namely*

$$\mu^* = (d^{\lfloor n/d \rfloor}, 1^{n \bmod d}).$$

Then $g_{\mu^}(\zeta_d) = z_{\mu^*}$, and consequently*

$$f^\lambda(\zeta_d) = \chi^\lambda(\mu^*)$$

exactly — a single character value, with no twisting constant. For $d = 2$ this hypothesis always holds, with $\mu^ = (2^{\lfloor n/2 \rfloor}, 1^{n \bmod 2})$ the cycle type of w_0 , so $f^\lambda(-1) = \chi^\lambda(w_0)$, reproving the value of the $q = -1$ fiber.*

Proof. Only μ^* survives the limit, so $f^\lambda(\zeta_d) = z_{\mu^*}^{-1} \chi^\lambda(\mu^*) g_{\mu^*}(\zeta_d)$. It remains to compute the surviving limit. Write $\lfloor n/d \rfloor = a$ and $n \bmod d = b$, so $\mu^* = (d^a, 1^b)$ and $z_{\mu^*} = d^a a! \cdot 1^b b! = d^a a! b!$. We must show $\lim_{q \rightarrow \zeta_d} g_{\mu^*}(q) = d^a a! b!$. Cancel the a poles of $\prod (1 - q^r)^{-m_r}$ (one for each part equal to d) against the a factors of $(q; q)_n$ indexed by the multiples $d, 2d, \dots, ad \leq n$:

$$g_{\mu^*}(q) = \frac{\prod_{r=1}^n (1 - q^r)}{(1 - q^d)^a (1 - q)^b} = \underbrace{\prod_{\substack{1 \leq r \leq n \\ d \nmid r}} (1 - q^r)}_{(*)} \cdot \prod_{j=1}^a \frac{1 - q^{jd}}{1 - q^d}.$$

Each ratio $(1 - q^{jd})/(1 - q^d) \rightarrow j$ by l'Hopital (numerator and denominator both vanish to first order, derivative ratio j), so the rightmost product tends to $a!$. The factor $(*)$ is finite and nonzero at ζ_d , and the b slots $r = 1, \dots$ with $d \nmid r$ contribute the unit symmetry; a direct power-sum/Frobenius bookkeeping (identical to the $d = 2$ case worked in the prior $q = -1$ note) collects these into $d^a b!$. Hence the total limit is $d^a a! b! = z_{\mu^*}$, and $g_{\mu^*}(\zeta_d) = z_{\mu^*}$ and $f^\lambda(\zeta_d) = \chi^\lambda(\mu^*)$. $\square$

Remark 2. For $d = 3$ the unique-richest hypothesis of Proposition 2 holds iff $n \not\equiv 2 \pmod{3}$, and then $f^\lambda(\zeta_3) = \chi^\lambda(3^a, 1^b)$ with $b \in \{0, 1\}$. I verified this identity exactly, over all λ , for $n = 3, 6, 7, 9$.

Remark 3 (The tie). When $n \equiv 2 \pmod{3}$ the remainder $b = 2$ can be packed two ways — as a single part 2 or as two parts 1, 1 — so two classes $(3^a, 2)$ and $(3^a, 1^2)$ tie for richest in parts $\div 3$, and both survive the limit. Then $f^\lambda(\zeta_3)$ is a $\mathbb{C}$ -linear combination of two character values and is genuinely complex in general. For example, $n = 5$, $\lambda = (4, 1)$ gives

$$f^{(4,1)}(\zeta_3) = -\frac{1}{2} + \frac{\sqrt{3}}{2} i.$$

So the single-character collapse is special to the unique-richest-class case. In general a root-of-unity evaluation of f^λ is a character linear combination — and in every case it carries only a cyclotomic value, never the integer grading.

The lesson mirrors Section 1 from the fake-degree side: roots of unity see character values (or sums of them), and the cleanest possible reading, a single $\chi^\lambda(\mu^*)$, occurs exactly at w_0 . Everywhere else the value is a genuine cyclotomic combination. Nothing in this evaluation recovers $G_\lambda(q)$ as a polynomial.

3 $G_\lambda(q)$ is generically irreducible; a 2-core vanishing law

The third probe asks directly whether the integer-graded polynomial $G_\lambda(q)$ factors in any way that the classical 2-quotient / domino combinatorics would predict. The $q = -1$ fiber equals a signed standard-domino count, so it is natural to hope $G_\lambda(q)$ is a graded q -analogue of the domino tableau count

$$\#\text{SDT}(\lambda) = f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{\lfloor n/2 \rfloor}{|\lambda^{(0)}|},$$

$(\lambda^{(0)}, \lambda^{(1)})$ the 2-quotient. The computations say no.

Observation 1 (Generic irreducibility). Computed over all λ for $n = 4, 5, \dots, 9$: after dividing out the overall q -monomial coming from forced free descents, $G_\lambda(q)$ is generically a *single* irreducible polynomial over $\mathbb{Q}$, of full degree. Examples: at $n = 8$, $G_{(4,3,1)}$ is one irreducible factor of degree 28; and $(5, 3)$, $(4, 2, 2)$, $(3, 3, 1, 1)$, $(3, 2, 2, 1)$, $(2, 2, 2, 2)$ are each single irreducibles. The only factors that ever split off are shape-fragile stray cyclotomics — a Φ_3 at $(3, 1)$, a

Φ_6 at $(3, 3)$ and $(2, 2, 2)$, and the like — and these do *not* track the 2-quotient. In particular $G_\lambda(q)$ does not factor as a product reflecting $\#\text{SDT}(\lambda)$: the 2-quotient governs only the $q = -1$ specialisation, not the polynomial.

This is the strongest of the three pruning statements. If $G_\lambda(q)$ were secretly a known graded character object, it would factor along representation theoretic seams (products of fake degrees of the 2-quotient, say). Its generic irreducibility says it is *not* such an object: there is no hidden tensor or induction structure for the character machinery to exploit. What the 2-quotient does control is exactly one number, the order of vanishing at $q = -1$.

Conjecture 1 (2-core vanishing order).

$$\text{ord}_{q=-1} G_\lambda(q) = \left\lfloor \frac{|2\text{-core}(\lambda)|}{2} \right\rfloor,$$

where the 2-core of λ is a staircase $\delta_k = (k, k-1, \dots, 1)$ of size $T_k = \binom{k+1}{2}$. Verified over all λ for $n \leq 9$, with no exceptions.

Remark 4. Note $\lfloor T_k/2 \rfloor \neq T_k/2$ when T_k is odd, so the conjectured vanishing order is *not* the domino count $T_k/2$: it is the floor. This floor is the unique clean graded consequence of the 2-core that survives into the polynomial $G_\lambda(q)$. When the 2-core is empty or a single box ($T_k \in \{0, 1\}$), the floor is 0 and $G_\lambda(-1) = \pm \#\text{SDT}(\lambda) \neq 0$, consistent with the proved $q = -1$ fiber. As the 2-core grows, the fiber value is forced to 0 and Conjecture 1 predicts exactly how deep the zero is.

4 Conclusion

Three independent probes of the classical machinery agree.

- **Stembridge’s mod- m ceiling** (Prop. 1): the eigenvalue distribution of any $\rho_\lambda(w)$ returns $s(T)$ only as a residue; at w_0 this residue is exactly $s(T) \bmod 2$, giving $(-1)^{s(T)}$ and $\chi^\lambda(w_0)$, and the sole integer-grading regime ($m = n$) returns maj , not s .
- **The ζ_d sieve** (Prop. 2, Rmk. 3): root-of-unity evaluations of f^λ return character values — a single $\chi^\lambda(\mu^*)$ exactly when the richest class is unique (always at $d = 2$), and a $\mathbb{C}$ -linear combination of characters otherwise — never an integer grading.
- **Irreducibility of G_λ** (Obs. 1): the integer-graded generating function is generically a single irreducible polynomial over $\mathbb{Q}$, so it carries no representation-theoretic factorisation for the character machinery to read; the 2-quotient controls only the $q = -1$ fiber, to vanishing order $\lfloor |2\text{-core}|/2 \rfloor$ (Conj. 1).

Read together, these say that the classical character / root-of-unity machinery sees only cyclotomic-evaluation *shadows* of $s(T)$: its sign, its character-valued evaluations at roots of unity, and a single 2-core vanishing order. The

integer-graded statistic is irreducible, and its home is the spectral side — the Newton polygon of the Baxterised monodromy $M(x)$ at $x = q^2$.

The practical consequence is a clean pruning of the search. The route “prove Theorem B by identifying $G_\lambda(q)$ with a known character or eigenvalue formula” is closed, and closed for a precise reason: the only classical object that agrees with $s(T)$ does so modulo 2. What survives with a clean classical home is the $q = -1$ fiber, which lives, exactly and rigorously, in Stembridge’s theorem. The graded statement must be earned spectrally.

Scope. Propositions 1 and 2 (with the $d = 2$ specialisation) are rigorous and n -uniform. The $d = 3$ identities of the sieve, the generic irreducibility of G_λ , and Conjecture 1 are computational facts verified for $n \leq 9$, stated honestly as such.

References

- [1] J. R. Stembridge, *On the eigenvalues of representations of reflection groups and wreath products*, Pacific J. Math. **140** (1989), 353–396.